Local Attention Routing for Wireless Sensor Networks with Reliability and Latency Constraints

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Abstract

Routing decisions in wireless sensor networks depend on energy, deadlines and uncertain forwarding behavior. We evaluate a local-attention constrained PPO router with class-specific service debt and Bayesian forwarding risk. A historical 30-seed comparison improved reliability but failed its joint superiority criterion. In a separate capacity-matched validation, attention increased critical-class delivered delay by 0.821 ms and reduced total violations by 1.522 percentage points; neither co-primary contrast was significant after Holm correction. Actual UDP feedback then connected received forwarding acknowledgments to frozen-policy decisions. In an exploratory shared-target fault diagnostic across 30 new fixtures, attention delivered 85.39% of packets and a capacity-matched MLP delivered 32.22%, a paired difference of 53.17 percentage points (95% CI 35.31–71.02). These fixed-state localhost results establish software integration under targeted relay failures. They do not establish live intrusion detection, realistic multihop wireless performance, absolute service-budget satisfaction or a general attention advantage.

Index Terms: Constrained reinforcement learning, Local attention, Risk-aware routing, Wireless sensor networks

I. Introduction

Wireless sensor networks (WSNs) serve distributed sensing, industrial monitoring, and smart infrastructure. Route selection is constrained by battery energy and must respond to changing radio links, queue conditions, and unreliable forwarding [1, 2, 3, 4, 5]. Traffic requirements also differ: a route suitable for delay-tolerant packets may be unsuitable for packets with short deadlines. A neighbor can look attractive in terms of expected transmissions or progress toward the sink yet become a poor choice when little deadline slack remains, its queue grows, or its forwarding record worsens.

Deep reinforcement learning (DRL) maps observations to sequential forwarding decisions through interaction with the network environment [6, 7, 8, 9]. Recent routing studies already combine value-based and policy-gradient

learning with attention, graph neural networks, federated learning, trust information, and action masks [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23]. We therefore examine a specific design question: how does a local candidate-set representation affect routing outcomes compared with a fixed-slot representation when both receive the same information and training budget?

LA-CPPO processes each candidate-neighbor feature row with a shared encoder. A query derived from the packet context scores the candidates, so a permutation of their rows produces the same permutation of actor logits. The critic uses a masked mean of the encoded rows. The action set contains at most eight candidates, and the model has no node-specific parameters. Training combines a scalar reward with six cost channels: total delay violation and loss for each of three traffic classes. A Beta forwarding-trust model and a smoothed queue-excess signal provide online risk context without exposing attack labels to the controller.

The architecture, reward coefficients, training budget, seed list, co-primary endpoints, multiplicity correction, guardrails, and decision rule were fixed before confirmatory outcomes were generated. The two learning arms share the state content, candidate set, mask, reward, costs, simulator settings, and evaluation-seed scheme. Their policy and value representations differ. Hidden width is matched, but parameter count is not; the comparison estimates the difference between the two specified implementations.

LA-CPPO reduced the critical-class total delay-violation rate and passed the delivery and energy guardrails, while Class-0 mean delivered delay remained numerically higher. Exploratory analysis associated the violation reduction mainly with fewer losses and a larger share of delivered three-hop packets. We then evaluated two residual-slack variants under a separate development protocol. Neither qualified for confirmation, and the reserved confirmatory seeds remained unused.

This study sets out a candidate-equivariant actor and invariant critic, a six-channel QoS formulation that uses online forwarding evidence, and a prespecified comparison with a state-matched MLP. The empirical finding is a reliability–latency trade-off: the mean-delay endpoint failed, and the later development refinement was unsuccessful. These outcomes limit the conclusions that can be drawn about the method’s routing performance.

II. Related work

A. Classical WSN routing and quality metrics

Earlier WSN research established energy-aware organization, link quality, and distributed forwarding with low overhead as central routing concerns [1, 2, 3, 4, 5]. LEACH uses clustering to spread the energy burden [3], while ETX expresses expected transmission requirements in terms of loss probability [4]. RPL later provided a standardized routing framework for low-power and lossy networks [5]. These approaches remain useful baselines and design

references. Their largely hand-designed rules, however, do not directly optimize heterogeneous application constraints as security and queue conditions change.

B. DRL routing, attention, and graph representations

Recent reviews show that DRL routing now spans value-based, policy-gradient, actor-critic, graph-based, and multi-agent methods across wired and wireless settings [10]. PPO has also entered QoS-aware ad hoc routing [17], while attention and hierarchical or multi-agent mechanisms appear in IoT-WSN routing frameworks [18, 23]. Local forwarder selection with DQN is used in opportunistic WSN routing [19], and graph-policy architectures are explicitly developed for routing generalization and permutation-related structural transfer [15, 16]. Consequently, PPO, attention, local-neighbor scoring, and permutation-aware learning cannot be claimed as individually novel primitives.

LA-CPPO shares candidate encoding and scoring across rows and pools the critic input symmetrically. A fixed-slot MLP receives the same serialized state. The historical comparison matches hidden width; the independent validation approximately matches total parameter count. Neither design matches every computational or architectural property.

C. Secure and trust-aware intelligent routing

Trust-aware routing was established before recent DRL methods. Yang et al. combined reinforcement learning with blockchain-based routing trust [11]. Subsequent studies use trust, cryptography, federated learning, or intrusion detection [21, 22, 24], and failure-risk-aware DQN routing incorporates reliability information into multi-hop decisions [14]. Comparisons between these approaches depend on what the router can observe. In LA-CPPO, attack labels remain inside the simulator; the policy receives forwarding observations, queue state, and the risk estimates derived from them.

D. Hybrid routing and clustering

Metaheuristic clustering and learned route selection are established hybrid designs. Wang et al. combine quantum-inspired Harris hawk optimization with DRL [12], and other IoT-WSN frameworks incorporate clustering or federated learning [13, 21, 23]. AF-v3 disables clustering, hierarchy masks, and the hard security shield to focus the primary comparison on local routing. A legacy ROA-based method is retained only as a descriptive baseline. Table I summarizes the methodological positioning.

III. System model and constrained routing problem

A. Network, radio, traffic, and threat model

The primary simulator uses N=100 static sensor nodes placed uniformly in a 150 m × 150 m region, with the sink at (75,75) m. Each node starts with 0.5 J. The nominal communication range is 50 m, packets are 4000 bits, and the physical data rate is 250 kbps. Radio energy follows first-order transmit/receive accounting with distance-dependent free-space and multipath terms. Link behavior also includes log-distance path loss, 3 dB shadowing, interference-dependent PDR, and up to three MAC retries. Queue pressure and interference persist across packet episodes and are refreshed at every sixth packet boundary rather than being reset for each packet.

Traffic is divided into three classes with generation probabilities 0.35, 0.45, and 0.20, deadlines of 90, 350, and 1500 ms, and target PDRs of 0.985, 0.96, and 0.90. Their delay-violation budgets are 0.015, 0.04, and 0.10. Class 0 is the critical low-latency class used for the co-primary endpoints. In the mixed-attack scenario, 25% of nodes are marked malicious for evaluation and are assigned blackhole, selective-drop, or flooding behavior internally. Selective dropping occurs with probability 0.65, while flooding raises local queue pressure. The policy observes neither the malicious-node labels nor the assigned attack types. Tables II and III summarize the configuration and threat-model visibility.

B. Virtual QoS debt

For each traffic class c, separate virtual queues track deadline-violation and packet-loss pressure. Let $u \in \{\mathrm{d}, \ell\}$ identify the deadline and loss channels, respectively. The index n counts completed packets of class c; only that class is updated at its packet termination:

$$Z_c^{(u)}(n+1) = \max\left\{0,\ Z_c^{(u)}(n) + C_c^{(u)}(n) - b_c^{(u)}\right\},\quad u \in \{\mathrm{d}, \mathrm{l}\}. \tag{1}$$

The binary event $C_c^{(u)}$ equals one for the corresponding terminal violation, as defined in Section III.D. Before entering the policy state, each debt is mapped to $\dfrac{Z_c^{(u)}}{1 + Z_c^{(u)}}$. The six transformed debts retain the ordering of accumulated QoS pressure while bounding the input scale. Debts from other classes remain unchanged at that termination. These virtual queues provide memory across packet episodes; they do not impose a hard per-packet delivery guarantee [25].

C. Label-free Bayesian forwarding risk

Every sensor starts with a Beta(1, 1) forwarding-trust prior. A forwarding observation at sensor i has weight $w_i = \mathrm{clip}(PDR_i^{\mathrm{exp}}, 0.05, 1)$. Success adds this weight to α_i; failure adds it to β_i. The posterior mean and standard deviation are $\mu_i = \dfrac{\alpha_i}{\alpha_i + \beta_i}$ and σ_i, respectively. For rate excess $x_i = \mathrm{clip}((q_i - 0.4)/0.6, 0, 1)$, the slow-state refresh applies $a_i \leftarrow 0.85\,a_i + 0.15\,x_i$. The mean risk and uncertainty-adjusted score are

$$\rho_i = 0.75\left(1 - \mu_i\right) + 0.25\,a_i, \tag{2}$$

$$\rho_i^{\mathrm{UCB}} = \mathrm{clip}\left(\rho_i + 1.645 \cdot 0.75\,\sigma_i, 0, 1\right). \tag{3}$$

Here q_i is normalized queue pressure and a_i is its smoothed excess-rate signal. The standard deviation follows the Beta variance $\sigma_i^2 = \dfrac{\alpha_i \beta_i}{\left(\alpha_i + \beta_i\right)^2\left(\alpha_i + \beta_i + 1\right)}$. Both risk quantities enter the candidate state. Forwarding evidence updates the current transmitting sensor after an attempted hop; attack identity never enters this estimator. The UCB term is an uncertainty-aware routing score, not a calibrated intrusion probability or a supervised IDS prediction.

D. Constrained routing objective

Let π_θ denote the next-hop policy. Consider a packet from class c with deadline D_c. Let D denote its observed end-to-end delay. The terminal costs are

$$C_c^{(d)} = 1\left\{\text{not delivered or delivered with } D > D_c\right\},$$
$$C_c^{(l)} = 1\left\{\text{not delivered}\right\}.$$

Loss therefore contributes to both channels. Continuing transitions have zero terminal costs. The six-element implementation vector stores the active class at positions 2c and 2c + 1. Delay-violation budgets are [0.015, 0.04, 0.10], and loss budgets are the complements of the target PDRs: [0.015, 0.04, 0.10]. The constrained scalar objective is

$$\begin{aligned} &\max_\theta \;\; J_R(\theta) = \mathrm{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \\ &\text{subject to} \;\; J_c^{(u)}(\theta) = \mathrm{E}_{\pi_\theta}\left[C_c^{(u)} \mid \text{class} = c\right] \le b_c^{(u)}, \\ &\qquad c \in \{0, 1, 2\}, \;\; u \in \{\mathrm{d}, \mathrm{l}\}. \end{aligned} \tag{4}$$

The expectation in each constraint is conditioned on the packet class; a global average over all traffic would dilute the class-specific budget. Here T is the number of forwarding decisions in the packet episode, and $\gamma = 0.99$. The training

routine estimates each class/channel rate from completed packets of that class in the current batch. Absent classes do not trigger multiplier updates. Section IV.E specifies the projected update and its order relative to PPO.

PPO uses reward and cost generalized advantages. With $\bar{A}^R$ denoting the reward advantage after the batch-standardization rule in Section IV.E, the combined advantage is

$$A_t^{\mathrm{CPPO}} = \bar{A}_t^R - \sum_{c=0}^{2} \left[\lambda_c^{(d)} A_{t,c}^{(d)} + \lambda_c^{(l)} A_{t,c}^{(l)} \right]. \tag{5}$$

Equation (4) states the routing objective and class-specific service targets. Equation (5) is the implemented PPO training surrogate: it combines standardized reward advantages with unstandardized, discounted cost advantages. The projected multipliers respond to undiscounted terminal violation rates. This procedure provides a penalty-based approximation to constrained learning; it does not establish exact feasibility of Eq. (4) [7, 8, 9, 25].

IV. Proposed LA-CPPO routing architecture

LA-CPPO maps a packet context and a bounded set of neighboring candidates to a next-hop action. Figure 1 shows how the persistent network state, class-specific QoS debt and online forwarding-risk estimates enter that decision. Shared candidate encoders produce the actor logits, while a symmetric masked pool supplies the reward and cost critics. Packet outcomes update the observed state online; batches of completed episodes drive the policy, critic and multiplier updates during training.

A. Packet context and candidate preselection

For each forwarding decision t, we represent the observation by a context vector $x \in \mathrm{R}^{13}$, a candidate matrix $H \in \mathrm{R}^{8 \times 10}$, and an eight-slot binary mask m. The context records the current energy fraction and queue pressure, sink distance relative to the deployment diagonal, remaining slack relative to the class deadline, a three-class one-hot vector, and six transformed QoS debts. We serialize the context and candidates into 93 values for batching, while retaining their distinct mathematical roles.

A candidate row records ten quantities: a sink indicator, residual-energy fraction, link PDR, normalized RSSI, queue pressure, normalized sink progress, mean posterior risk, risk-UCB, expected retries, and expected hop delay. We clip energy fractions and RSSI to [0, 1], with RSSI transformed as clip((RSSI + 110)/60, 0, 1). Sink progress is divided by the radio range and clipped to [−1, 1]. Expected retries use min(1/max(PDR, 10^{-3}) − 1, 4)/4, and the delay feature is clip(expected hop delay/class deadline, 0, 2)/2. For a sink candidate, energy is one and queue/risk entries are zero.

These transforms use fixed physical scales or ratios; their limits are not fitted through dataset-dependent min–max normalization.

Preselection considers alive, reachable neighbors that have not yet been visited, adding the sink when it is reachable. Candidate j is ranked using

$$s_j = 1.25\,p_j + 0.75\,\mathrm{PDR}_j + 0.25\,e_j - 0.55\,\rho_j^{\mathrm{UCB}} - 0.20\,q_j. \tag{6}$$

In this score, p_j measures the reduction in sink distance divided by the radio range, e_j is the residual-energy fraction, and q_j is candidate queue pressure. We retain the eight highest-scoring candidates and zero-pad and mask any missing rows. The ranking determines which next-hop options are available; the actor makes the final choice. The mask excludes unavailable slots after physical and visited-node filtering. The environment enforces the 22-hop limit at termination, without an additional learned or security mask.

B. Shared local-attention actor

The context encoder f_x and shared candidate encoder f_h each contain two affine layers with tanh activations and hidden width d = 96. The context query and candidate keys use bias-free linear projections. A candidate-wise bias network has widths 96 → 48 → 1 with a tanh hidden activation. Candidate j receives the unmasked logit

$$l_j = \frac{\left\langle W_q f_x(\mathrm{x}),\, W_k f_h\left(h_j\right)\right\rangle}{\sqrt{d}} + b_{\mathrm{act}}\left(f_h\left(h_j\right)\right). \tag{7}$$

All rows share the same encoding, projection and bias parameters. Masked slots receive logit -10^9 before softmax. During training, the action is sampled from this categorical distribution; evaluation chooses its argmax. The bias function b_{act} is distinct from the QoS budgets $b_c^{(u)}$. No node identifier, cluster role, attack label or global topology embedding enters the actor.

C. Invariant critics and representation properties

The critics use the masked mean g of the encoded candidate rows:

$$g(\mathrm{H},\mathrm{m}) = \frac{\sum_{\mathrm{j}=1}^{K} m_j f_h\left(h_j\right)}{\max\left(1,\, \sum_{\mathrm{j}=1}^{K} m_j\right)}.$$

We concatenate $f_x(\mathrm{x})$ with g to form a 192-dimensional input to the critic's 96-unit tanh layer. Separate linear heads produce one reward value and six cost values. Applying the same permutation matrix P to the candidate rows and mask gives the following identities through shared row operations and symmetric pooling:

$$L(\mathrm{x}, \mathrm{PH}, \mathrm{Pm}) = \mathrm{P}\, L(\mathrm{x}, \mathrm{H}, \mathrm{m}),$$
$$V(\mathrm{x}, \mathrm{PH}, \mathrm{Pm}) = V(\mathrm{x}, \mathrm{H}, \mathrm{m}).$$

Reordering the selected candidates therefore reorders the logits and masked action probabilities in the same way. The reward value and all six cost values remain unchanged. These identities apply after the candidate set has been selected. Ties can still affect node choice: deterministic argmax tie-breaking and selection at the top-K boundary need not preserve the chosen node after reordering. The representation property alone establishes neither network-size transfer nor superiority at equal capacity [26, 27]. Encoding and scoring grow linearly with K at fixed hidden width; the model does not use all-pairs candidate self-attention.

D. Hop reward and terminal-event handling

Once an attempted hop passes action validation, Δd_t denotes progress toward the sink in metres, R = 50 m is the radio range, n_t is the number of transmission attempts, and d_t is the hop distance. The base reward is

$$r_t^{\mathrm{hop}} = 0.90\ \mathrm{clip}\left(\frac{\Delta d_t}{R}, -1, 1\right) - 0.20\, q_t - 0.35\, \frac{n_t E_{\mathrm{tx}}(d_t)}{E_0} - 0.45\ 1_{\{a_t \neq \mathrm{sink}\}}\, \rho_{a_t}^{\mathrm{UCB}} - 0.04 \max(n_t - 1, 0). \tag{8}$$

The selected target is a_t, the transmitting node's queue pressure is q_t, and E_0 = 0.5 J. The term $\mathrm{E}_{\mathrm{tx}}(d_t)$ gives the transmit energy for one packet attempt. Equation (8) charges the transmitter's attempts; the simulator tracks receiver energy separately. The indicator omits the candidate-risk penalty on sink-directed hops. Sink risk is zero, so no sensor-risk lookup is needed for the sink.

Before an attempted hop, the environment checks the selected slot and its range. An invalid, empty, or out-of-range action ends the episode with reward −3.0. For other actions, Eq. (8) applies even if forwarding fails. Failed forwarding adds −2.0. Successful delivery to the sink adds +2.5, with a deduction of 0.8 for late delivery. After a successful relay hop, a loop, the hop limit, or a dead end adds −2.0 and ends the episode. Continuing hops receive no terminal increment. The active class's delay/loss costs and QoS debts update at packet termination, whereas forwarding-risk evidence updates after every attempted hop. We fixed all coefficients before observing the confirmatory outcomes.

Algorithm 1 summarizes a forwarding decision and the resulting transition.

Algorithm 1. One LA-CPPO forwarding decision

Input: Node i, class c, visited set S, policy θ, persistent network and QoS state.

Output: Next-hop transition (o, a, r, C, o′, done).

1. N ← alive, reachable neighbors of i outside S
2. if sink is reachable then N ← N ∪ {sink}
3. for each j ∈ N do
4. $h_j \leftarrow$ CandidateFeatures(i, j, c); $s_j \leftarrow$ Score(i, j) [Eq. (6)]
5. end for
6. H, m ← TopKAndPad({(s_j, h_j)}, K = 8)
7. x ← PacketContext(i, c, remaining slack, six QoS debts)
8. ℓ ← MaskedActor(x, H, m) [Eq. (7)]
9. a ← Sample(softmax(ℓ)) in training; argmax(ℓ) in evaluation
10. o′, r, C, done ← EnvironmentStep(a) [Eq. (8); terminal rules]
11. return ((x, H, m), a, r, C, o′, done)

E. Constrained PPO training and matched comparator

Each seed uses eight training updates, 50 complete packet episodes per update, and three full-batch PPO epochs per update. Reward and all six cost advantages use GAE with γ = 0.99 and λGAE = 0.95. If the reward-advantage standard deviation exceeds 10^{-6}, its batch mean is subtracted and it is divided by its standard deviation plus 10^{-8}; otherwise it is retained. Cost advantages are not standardized. Reward and cost return targets are computed before reward-advantage standardization [7, 8].

After collecting a batch, $\hat{J}_c^{(u)}$ is the average terminal cost among packets of class c in that batch. The multiplier starts at zero and is updated before the three PPO epochs:

$$\lambda_c^{(u)} \leftarrow \Pi_{[0,\,50]}\left[\lambda_c^{(u)} + \eta_\lambda\left(\hat{J}_c^{(u)} - b_c^{(u)}\right)\right], \quad \eta_\lambda = 0.03. \tag{9}$$

Equation (9) is applied only to classes represented in the batch; absent classes retain their multipliers. Projection clips both the lower and upper bounds. The resulting multipliers remain fixed across the three PPO epochs and are used in Eq. (5).

For policy ratio $R_t(\theta) = \dfrac{\pi_\theta\left(a_t \mid o_t\right)}{\pi_{\theta_{\text{old}}}\left(a_t \mid o_t\right)}$, the clipped policy objective is

$$L^{\text{clip}}(\theta) = \hat{\mathrm{E}}_t\left[\min\left(R_t(\theta)\,A_t^{\text{CPPO}},\ \text{clip}\left(R_t(\theta),\,0.8,\,1.2\right)A_t^{\text{CPPO}}\right)\right]. \tag{10}$$

The joint loss minimizes the negative of Eq. (10), subtracts 0.01 times mean policy entropy, and adds the reward and cost mean-squared-error terms. Each value error is first multiplied by 0.5 and then weighted by a further 0.5 in the joint loss; the cost error averages over transitions and all six channels. Adam uses learning rate 3×10^{-4}, and the joint gradient norm is clipped at 1.0. Algorithm 2 states the implemented order.

Algorithm 2. Lagrangian PPO training

Input: Actor/critic parameters θ, ψ; class budgets b; frozen training configuration.

Output: Trained parameters θ, ψ and final multipliers λ.

$\lambda \leftarrow 0$
for update = 1, …, 8 do
    B, $\hat{J}$, active $\leftarrow$ Collect(50 packet episodes; Algorithm 1)
    A^R, G^R, A^c, $G^c \leftarrow$ GAE(B; $\gamma = 0.99$, $\lambda_{GAE} = 0.95$)
    $\bar{A}^R \leftarrow$ StandardizeIfVarying(A^R); retain the raw cost advantages
    $\lambda \leftarrow$ ProjectActive(λ, $\hat{J} - b$; $\eta_\lambda = 0.03$, bounds [0, 50]) [Eq. (9)]
    $A^{cPPO} \leftarrow$ CombinedAdvantage($\bar{A}^R$, A^c, λ) [Eq. (5)]
    for epoch = 1, 2, 3 do
        Form the clipped policy, entropy and value losses [Eq. (10)]
        Backpropagate; clip gradient norm at 1.0; take one Adam step
    end for
end for
return θ, ψ, λ

Evaluation uses 240 packet episodes in a newly initialized environment with separate seed offsets, the trained model weights, and deterministic actions. Forwarding evidence and QoS debts continue to update online during evaluation; the policy and critic weights remain frozen. The fixed-slot MLP receives the same 93 observation values, candidate mask, reward/cost signals, optimizer settings and training budget. Its two 96-unit tanh layers process the flattened observation, followed by actor and value heads.

The historical comparison was not parameter-count matched: attention had 63,368 parameters and the same-width MLP had 19,791. The separate new validation approximately matches total parameter count and is reported in Section IX. Neither design isolates architecture at equal computation cost.

V. Confirmatory experimental methodology

A. Prespecified hypothesis, estimands, and seeds

The confirmatory hypothesis required local attention to outperform the identical-state MLP on both Class-0 endpoints. Differences are reported as attention minus MLP, so negative values are favorable. Thirty fresh seeds were reserved before AF-v3 outcomes were generated and were not used for structural qualification, smoke tests, or development. Training-environment, Torch, NumPy, and evaluation streams use fixed seed offsets so that the evaluation stream does not simply continue the training stream. All 30 paired seeds were retained.

We estimate the across-seed paired differences in Class-0 mean delivered delay and Class-0 total delay-violation rate. A total violation includes either non-delivery or arrival after the 90 ms deadline. Using both endpoints makes the selection effect in delivered-packet latency visible: the mean can look lower when difficult packets are lost and never enter the delay calculation.

B. Statistical analysis and decision rule

For each co-primary endpoint, the prespecified analysis reports the paired mean difference, a two-sided 95% confidence interval, a paired t-test, Cohen dz, and a Wilcoxon signed-rank sensitivity analysis. We apply Holm adjustment to the two paired t-test p-values. Delivery and energy are assessed with two one-sided 95% non-inferiority checks: the lower confidence bound for the overall delivery-rate difference must exceed -0.01, and the upper confidence bound for the geometric energy-per-delivered-packet ratio must not exceed 1.05. A strict PASS requires favorable differences on both co-primary endpoints, both Holm-adjusted p-values below 0.05, and both guardrails to pass. It also requires all matched seeds, apart from a documented whole-seed integrity failure, and no protocol deviation.

C. Simulation data and evaluation samples

We generated the training and evaluation data as simulator trajectories. Each learning arm used 400 training packet episodes and 240 evaluation episodes per seed, giving 12,000 and 7,200 episodes across the 30 seeds. During evaluation, actions are deterministic but the environment remains stochastic. The realized Class-0, Class-1, and Class-2 counts are 2,514, 3,283, and 1,403 for attention and 2,527, 3,230, and 1,443 for the MLP. Random class draws and policy-dependent consumption of the random stream account for these differences. Training and evaluation start from separately initialized environments. There is no offline train–test split or supervised intrusion-detection dataset.

D. Random-stream pairing

Both policies use the same seed to initialize a network. Subsequent external events are not replayed from a policy-independent schedule, however. Source and traffic-class draws use the same environment random stream as some transition events, so different actions and episode lengths can alter later draws. The pairing therefore aligns the initial conditions without ensuring identical traffic, radio, or attack events thereafter. Separate random streams or replay of an exogenous event schedule would improve a future replication.

E. Reproducibility and protocol integrity

We fixed the architecture and analysis plan before evaluating efficacy. The AF-v3 E1 record is GitHub Actions run 34178527532 at source commit 0f60eb5; full identifiers and artifact digests are listed in the provenance manifest. An earlier execution completed the scientific computation but failed while writing provenance paths. The technical rerun preserved the architecture, seeds, coefficients, budgets, endpoints, and statistical rules.

F. Descriptive training diagnostics

Figure 2 follows training across the 30 seeds. Between archived update 0 and update 7, mean episode reward rose from 0.056 to 1.729 for attention and from −0.090 to 1.866 for the MLP. Mean training delivery rose from 0.534 to 0.749 and from 0.501 to 0.759, respectively. These curves show progress within the allocated budget, but they do not establish convergence or optimality. The shading represents pointwise 95% t intervals for the across-seed mean.

VI. Results

A. Co-primary confirmatory endpoints

Figures 3 and 4 show seedwise outcomes for both policies on the co-primary endpoints. Mean delivered delay is numerically higher under attention, and its confidence interval includes zero. It therefore meets neither the directional requirement nor the multiplicity-adjusted significance requirement. The Class-0 total violation rate falls by 0.03125 absolute, with its confidence interval entirely below zero and Holm-adjusted p=0.007703. The paired standardized effect is moderate (dz=-0.574). Both endpoints had to succeed, so these results do not confirm the prespecified superiority hypothesis. Table IV reports the paired inferential results.

B. Non-inferiority guardrails

Attention passed both non-inferiority checks. Overall delivery was higher on average, and energy per delivered packet stayed within the prespecified margin. The failed mean-delay endpoint nevertheless means that the joint success criterion was not met. Table V reports the estimates and frozen margins.

C. Descriptive baselines

Table VI reports static energy-risk-ETX and periodic ROA energy-risk-ETX routing as descriptive baselines. Both use the AF-v1 configuration, whose clustering, hierarchy, and shield settings differ from the primary experiment. These comparisons provide context but cannot isolate the effect of policy representation. They are therefore outside the confirmatory endpoint family and strict decision rule.

VII. Exploratory mechanism analysis

The analyses in this section were carried out after the confirmatory decision and are treated as exploratory. They are used to understand why the two primary endpoints moved differently; they do not modify the pre-specified verdict or introduce a new success criterion.

A. Exact decomposition of total violation

For Class 0, a total violation consists of one of two mutually exclusive outcomes: the packet is not delivered, or it is delivered after the 90 ms deadline. The attention-minus-MLP difference is -0.027483 for Class-0 loss and -0.003771 for the delivered-late fraction. These components sum to -0.031254, exactly reproducing the confirmatory total-violation difference. In magnitude, about 87.9% of the observed improvement therefore comes from fewer losses rather than from fewer late deliveries. Table VII summarizes this decomposition.

B. Route-length composition and survivor bias

Figure 5 shows the route-length composition. Among delivered Class-0 packets, attention increases the per-seed mean hop count by about 0.0648 and the three-hop share by about 0.0458. Within a fixed hop count, pooled delivered delays are not higher for attention: 30.083 versus 30.136 ms at one hop, 60.926 versus 62.098 ms at two hops, and 90.196 versus 96.368 ms at three hops. The larger difference is in route composition: attention delivers 186 three-hop Class-0 packets, whereas the MLP delivers 75.

The pooled delivered means are 52.19 ms for attention and 51.26 ms for the MLP. Reweighting attention's within-hop means with the MLP hop proportions gives 50.35 ms. This calculation is consistent with route composition contributing to the observed delay difference. It does not establish that attention causally improves delay within comparable packets: the delivered populations differ, and the reweighting is neither randomized nor a mediation analysis.

C. Security-drop pattern

The overall security-drop rate is 0.0308 for attention and 0.0649 for the MLP, and the paired Class-0 delivery difference is approximately +0.0275. These descriptive results are consistent with greater use of lower-risk forwarding opportunities. The logs do not identify which individual packets would have survived under both policies, so they cannot establish a causal path from risk scoring to route length and delay.

VIII. Development-only residual-slack sensitivity

After AF-v3 E1, we fixed a separate AF-v4 development protocol and evaluated two residual-slack attention variants against a state-matched slack-aware MLP on 20 fresh seeds. Neither met the selection criteria. The candidate-minus-MLP Class-0 mean-delay differences were +3.226 ms for the slack-feature model and +1.628 ms for the slack-monotone model, although both delivery and energy guardrails passed. No candidate was selected, and the 30 reserved AF-v4 confirmatory seeds remained unused. Appendix A reports the criteria, intervals, and decision.

IX. Independent reviewer-led validation

This validation was specified separately after the historical AF-v3 and AF-v4 programs. The original gates and outcomes remain unchanged. Thirty new evaluation seeds, 41001–41030, were fixed before the new outcomes were generated. Each arm received eight updates of 50 training episodes and three PPO epochs per update. Trained weights were evaluated for 240 episodes in each setting and retained as checkpoints.

The attention model contains 63,368 parameters. The nearest-size fixed-slot MLP contains 63,554. Its width was chosen from parameter count alone. This controls total parameter count approximately; compute, layer structure and inductive bias remain different. Two additional attention arms remove risk from candidate features, preselection and reward, or observe only 80% of forwarding outcomes. The latter is a missing-feedback sensitivity analysis, not a live-traffic implementation.

Separate seeded generators govern traffic selection, queue dynamics, attack transitions, radio draws and feedback omission. Radio consumption therefore does not alter later traffic-class draws. Policy-dependent radio and attack draws remain possible, and source eligibility can diverge after node death. The experiment does not replay an identical per-edge external-event schedule.

The nominal comparison family consists of the six paired tests formed by three comparators and two critical-class endpoints. Holm correction is applied jointly to these six tests. Paired 95% intervals are unadjusted. Transfer evaluations retain the trained weights and change only the node count to 200 in the same area or the malicious-node ratio to 10% or 40%. These secondary results measure sensitivity; increasing density is not a density-preserving scalability test. Table IX summarizes the nominal and transfer outcomes.

Against the capacity-matched MLP, attention changed mean delivered delay by +0.8205 ms (95% CI -0.1212 to +1.7623; Holm p=0.3409). Total violations changed by -1.5223 percentage points (95% CI -2.7538 to -0.2909; Holm

p=0.08581). Table VIII reports all nominal comparisons, and Fig. 6 shows their effect intervals. Unfavorable and inconclusive contrasts are retained.

In the nominal simulation study, forwarding observations come from simulator outcomes. The separate UDP diagnostics below validate packet/feedback software integration but retain synthetic link features and omit realistic radio overhead.

A. Packet feedback and frozen policy integration

The software feedback adapter sends an actual UDP datagram to the selected relay and accepts an acknowledgment only from the intended endpoint with matching sequence and nonce. A separate sink socket witnesses downstream receipt before the relay acknowledges forwarding. The acknowledgment or timeout updates the Bayesian posterior before the next observation and policy inference. No attack label or fault-target identifier is supplied to the policy. Nonces prevent packet confusion but do not authenticate peers. Expected link PDR remains model-derived. In the simulator, a hop outcome is credited to the current transmitting node; in the UDP diagnostic, a downstream receipt is credited to the selected relay after its relay-to-sink forwarding. The observation point differs, so the diagnostic is an interface-transfer test rather than a bitwise reproduction of simulator feedback.

A preliminary periodic-fault fixture changed policy probabilities but produced no next-hop changes in 360 attempts. A subsequent persistent initial-relay fault diagnostic used all 30 frozen attention checkpoints and 30 fresh fixtures. Twenty-five fixtures selected an alternative relay; five did not. In the 25 first-switch decisions, the faulty relay remained a candidate and was unmasked. These exploratory diagnostics motivated the separately specified capacity comparison below; they are not confirmatory trials.

B. Shared target capacity comparison

Thirty fresh topology seeds, 73001–73030, were paired with the existing attention and capacity-MLP checkpoints, 41001–41030. Each fixture used two shared fault targets: the relay initially preferred by attention and the relay initially preferred by the MLP. Both controllers faced both targets for 30 attempts each. No model was retrained or selected after the outcomes. Source position, energy, queues and QoS state stayed fixed within each run; only forwarding-risk posteriors evolved. The relay fault was persistent and targeted, rather than a sample from a random malicious-node distribution.

The two target choices coincided in 26 fixtures. We averaged their outcomes within fixture before paired inference, retaining n = 30 rather than treating the 120 executed runs as independent experiments. Both policies initially chose the faulty relay in 56 of their 60 scenarios. Mean sink delivery was 85.39% for attention and 32.22% for the capacity-matched MLP. The paired difference was 53.17 percentage points (95% Student-t interval, 35.31–71.02; two-sided

paired $p = 1.246 \times 10^{-6}$). Attention performed better in 21 fixtures, tied in eight and worse in one. Table X and Fig. 7 report these diagnostic outcomes.

C. Verification and interpretation limits

Across 3,600 attempts, the sink witnessed 2,117 deliveries and 1,483 attempts timed out. The adapter and sink witness agreed on every recorded outcome. A separate verifier reconstructed the observations, risk posteriors, checkpoint hashes, policy probabilities and selected actions for all records. All 47 repository tests passed. This verifies internal software consistency; it is not an external packet-capture or authenticated observer audit.

The UDP payload is a seven-byte fixture message plus JSON metadata, not the simulator's 4,000-bit application packet. Relays communicate directly with the localhost sink regardless of radio reachability. The 20 ms timeout is a harness setting, not a measured wireless deadline. MAC contention, fading, sleep currents, acknowledgment energy and distributed authentication are absent. The positive diagnostic therefore supports a feedback-responsive frozen routing pipeline in this targeted fixture, while the earlier nominal co-primary results remain unchanged.

X. Discussion

A. Reliability and delivered-packet delay

In the tested 100-node mixed-attack setting, LA-CPPO lowered Class-0 total violations relative to the fixed-slot MLP under the same training budget. Both the Holm-adjusted paired t-test and the Wilcoxon sensitivity analysis supported this endpoint, and the delivery and energy guardrails passed. Because non-delivery counts as a violation, the improvement chiefly reflects a useful reliability gain.

Passing the relative guardrails does not imply that the absolute class budgets were met. The across-seed Class-0 total-violation and loss rates under attention were 0.1242 and 0.0480, respectively, against budgets of 0.015 for each channel. Class-1 rates also exceeded their 0.04 budgets. Table XI reports all six descriptive rates. The finite-budget Lagrangian training procedure therefore did not establish empirical constraint satisfaction in the evaluated setting.

Attention had a numerically higher mean delivered delay, with a confidence interval spanning zero, and the joint success criterion failed. Mean delay among delivered packets can favor a policy that loses the more difficult traffic. The reliability gain must therefore be read alongside this unsuccessful delay endpoint.

B. Representation-level interpretation

The fixed-slot MLP can use deterministic candidate rank as a cue. The shared attention model has no equivalent slot-specific parameters. Route-composition and security-drop patterns suggest that the two implementations use candidate information differently, but unequal parameter counts prevent an attribution to attention at equal capacity.

The unsuccessful AF-v4 refinement also shows that the tested residual-slack additions did not remove the delay trade-off.

C. Implications for evaluation of learned routing

Loss, delivered-packet delay, and total violations need to be reported together when evaluating learned routing. Holding state content and masks constant helps interpret an architectural comparison, while prespecified guardrails state how much deterioration in delivery or energy is acceptable. Effect sizes, confidence intervals, multiplicity control, and the reporting of failed endpoints make the limits of a local performance result clear.

D. Scope relative to current literature

Recent work combines DRL, attention, graph representations, trust, and federated learning in many routing settings [12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23]. LA-CPPO's contribution lies in its explicit local candidate-set formulation and its evaluation: an equivariant actor and invariant critic conditioned on QoS debt and online posterior risk, compared with a state-matched MLP under a fixed protocol. The present experiment establishes neither priority over all related architectures nor state-of-the-art performance.

XI. Study limitations

The confirmatory evidence comes entirely from simulation and from a single nominal setting of 100 nodes in a 150 m × 150 m area. It should not be read as evidence of field-deployment performance.

Historical confirmation uses a 25% mixed-attack setting. The separate validation changes attack prevalence to 10% and 40% but retains the same attack mechanisms. Robustness to unseen attack types, collusion, adaptive evasion, Sybil attacks and wormholes remains unknown.

Historical AF-v3 transfer and overhead experiments were not run after its failed gate. The separate new validation reports limited frozen-policy sensitivity to node count and attack prevalence. No hardware benchmark or complete deployment-overhead evaluation has been performed.

We use the Beta model to supply forwarding-risk context. It is not a validated intrusion detector: its uncertainty score has not been calibrated against real attack observations, and this study does not estimate intrusion-detection accuracy.

AF-v4 is development evidence only. It is kept separate from the AF-v3 confirmatory sample and is not used to redefine the primary endpoint or enlarge the inferential sample.

Candidate preselection still relies on a hand-designed score. Both learning arms receive the same ranked candidate set, which keeps the primary comparison aligned, but this study does not isolate the effect of the preselection rule itself.

The seed-paired experiment aligns initial stochastic conditions but does not replay an identical exogenous event trace across policies. Policy-dependent random-number consumption can therefore make later traffic, radio, or attack events diverge. This limits a strict common-random-number causal interpretation and should be addressed with split random streams or event replay in future replication.

The historical MLP is matched on state content and training budget rather than parameter count. The new comparison controls total capacity approximately; neither comparison establishes an isolated effect of representation at equal computation cost.

XII. Reproducibility record

The public package contains the current source, configurations, historical AF-v3 outputs, new validation data and checkpoints, analysis scripts and a file checksum manifest. Historical source commits and AF-v4 records are identified by links to their original provenance. Exact historical training requires the original source commit and environment; the current validation is not a bitwise reproduction claim.

XIII. Conclusion

LA-CPPO combines shared candidate scoring, invariant critics, service debt and forwarding-risk estimates. The historical joint superiority gate failed, and neither nominal capacity-matched co-primary contrast was significant after Holm correction. Risk removal reduced reliability. A separate targeted UDP diagnostic showed higher delivery for attention than the capacity MLP and verified packet-to-feedback-to-policy integration. Its fixed-state localhost design does not establish physical wireless effectiveness, live IDS accuracy or absolute constraint satisfaction. The evidence supports further deployment-relevant evaluation rather than a universal attention advantage.

Appendix A Historical development and constraint rates

The AF-v4 development output retained two unsuccessful candidates. Their mean-delay differences against the slack-aware MLP were +3.225549 ms and +1.628427 ms. Neither passed the frozen development selection gate; no candidate was selected. Raw development comparisons, episode logs and selection_decision.json are supplied under data/historical_af_v4. These outcomes are not pooled with either 30-seed study.

Declarations

Conflict of interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding: No external funding is declared for this study.

Ethics approval: Not applicable. This simulation study involves no human participants, animals, clinical data, or identifiable personal data.

Data and code availability: Source code, configurations, historical AF-v3 raw outputs, new validation trajectories, seed summaries, trained checkpoints and analysis scripts are available at https://github.com/DrShivamBhardwaj/JICCE_LA_CPPO_Reproducibility. Historical and new evidence are separated, and the original failed joint criterion is retained.

Credit author statement: Shivam Bhardwaj: Conceptualization, methodology, software, investigation, formal analysis, visualization, writing - original draft. Jitendra Nath Srivastava: Supervision, validation, writing - review and editing. Akash Sanghi: Supervision, validation, writing - review and editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used OpenAI tools to assist with language editing and preparation of code and analysis. They reviewed the methods, experimental outputs and manuscript and take responsibility for the final content.

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Table I. Positioning against representative recent intelligent-routing studies.

Study	Setting	Learning structure	Primary emphasis	Representation/design	Evaluation design reported
Wang et al. [12]	WSN	QHHO + DRL	Energy/lifetime	Cluster-dependent hybrid	Method positioning; no cross-paper numerical comparison
Manogaran et al. [13]	IoT-WSN	Adaptive DDQN + FL	Energy/routing	Learned routing; federated	Method positioning; no cross-paper numerical comparison
Tao et al. [14]	LPWAN	DQN	Failure risk	Risk-aware multi-hop	Method positioning; no cross-paper numerical comparison
Song et al. [15]	General networks	DRL + graph transformer	QoS/ generalization	Graph-structured policy	Method positioning; no cross-paper numerical comparison
Wigmore & Modiano [16]	Queueing networks	GNN policy learning	Generalization	Permutation-aware graph policy	Method positioning; no cross-paper numerical comparison
Venkatesan et al. [17]	MANET	PPO + slicing	QoS	Local topology/queue state	Method positioning; no cross-paper numerical comparison
Betam et al. [18]	IoT-WSN	Hybrid DRL + attention	Energy/delay/ throughput	Attention + hierarchy	Method positioning; no cross-paper numerical comparison
Iqbal & Siddiqui [20]	WSN	Feasibility-aware RL	Hop reliability/trust	Hard feasibility mask	Method positioning; no cross-paper numerical comparison
This work	IoT-WSN	Local attention + Lagrangian PPO	QoS + online risk	Candidate equivariance + invariant critic	30 paired seeds; frozen endpoints; Holm + NI guardrails

Table II. Frozen primary simulator and training configuration.

Parameter	Frozen value
Network / deployment	100 nodes; 150 m × 150 m; sink at (75,75) m
Initial energy / radio range	0.5 J per node / 50 m
Packet / data rate	4000 bits / 250 kbps
MAC retries	3 retries; at most 4 transmission attempts per hop
Traffic-class probabilities	[0.35, 0.45, 0.20]
Deadlines (ms)	[90, 350, 1500]
Target PDR	[0.985, 0.96, 0.90]
Delay-violation budgets	[0.015, 0.04, 0.10]
Malicious-node ratio	0.25
Selective-drop probability	0.65
Candidate limit	K=8
Hidden width / learning rate	96 / 3×10^{-4}
PPO clip / entropy	0.20 / 0.01
Discount factor γ / GAE λ	0.99 / 0.95
Lagrange learning rate	0.03
Training per seed	8 updates x 50 episodes x 3 PPO epochs
Evaluation per seed	240 episodes; deterministic policy
Clustering / hierarchy / shield	disabled / disabled / disabled

Table III. Threat-model behavior and controller visibility.

Component	Simulator behavior	Controller-visible information
Blackhole	Malicious forwarder drops forwarding attempts	No attack label; only posterior forwarding evidence
Selective drop	Drops with frozen probability 0.65	No attack label; posterior evidence only
Flooding	Adds queue pressure / congestion	Queue and link effects are observable; attack type is not
Attack truth	Evaluator-only malicious-node metadata	Never exposed to policy or risk estimator



Fig. 1. Routing architecture and constrained training procedure for LA-CPPO.

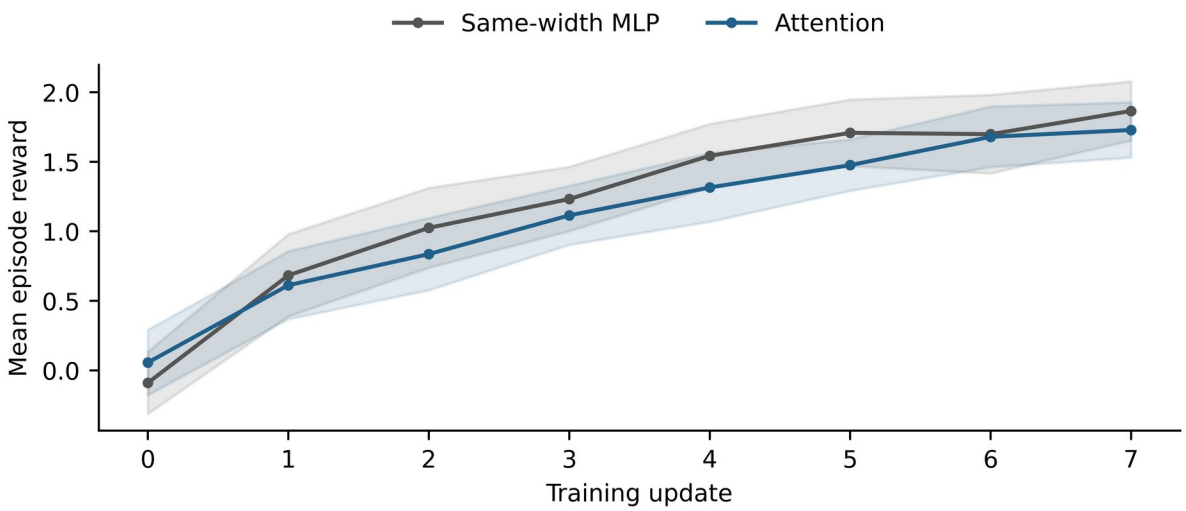


Fig. 2. Training reward across 30 historical seeds. Shading shows pointwise 95% t intervals.

Table IV. AF-v3 E1 co-primary confirmatory results (n=30 matched seeds).

Endpoint	Mean diff.	95% CI	Holm p	Wilcoxon p	Cohen d_z	Direction
Class-0 mean delivered delay	+1.0158 ms	[-0.5039, +2.5354]	0.1821	0.1981	+0.2496	Not favorable
Class-0 total delay-violation rate	-0.03125	[-0.05160, -0.01091]	0.007703	0.003475	-0.5736	Favorable

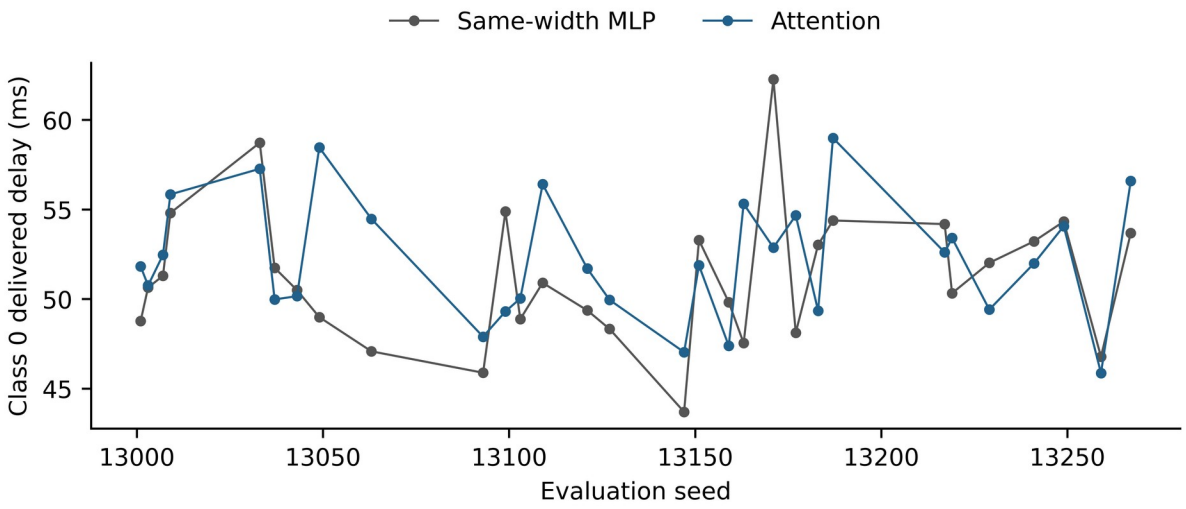


Fig. 3. Class-0 mean delivered delay for each paired seed.

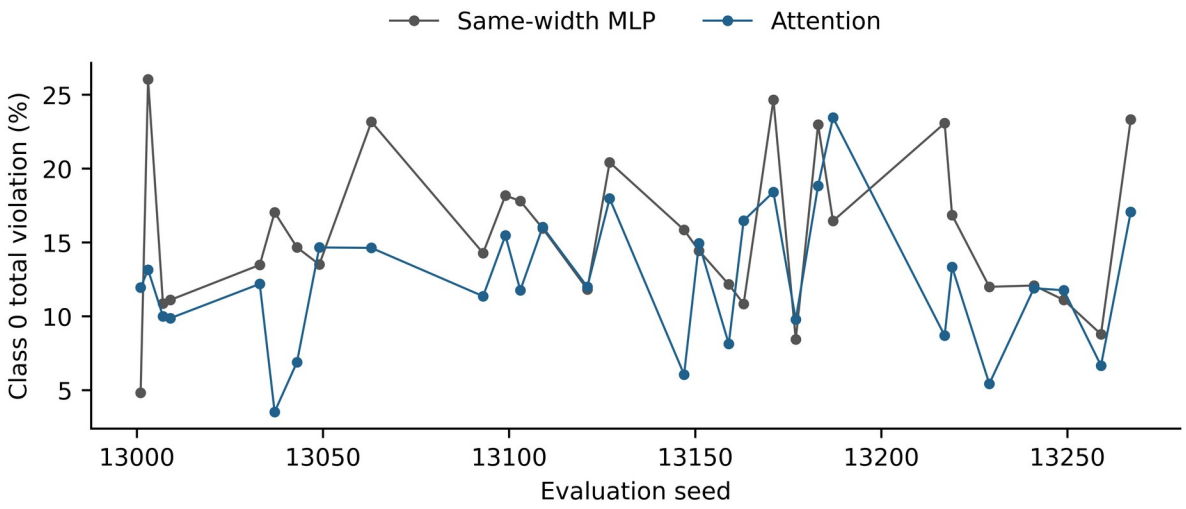


Fig. 4. Paired-seed Class-0 total delay-violation rate.

Table V. Prespecified guardrails for delivery and energy.

Guardrail	Estimate	Confidence bound	Frozen margin	Result
Overall delivery rate	+0.03153 absolute	one-sided 95% LB = +0.02158	>-0.01	PASS
Energy per delivered packet	GMR = 0.97180	one-sided 95% upper = 1.00040	≤1.05	PASS

Table VI. Mean evaluation metrics across 30 seeds; baseline rows are descriptive only.

Policy	Delivery	C0 delay (ms)	C0 violation	Energy/deliv. (mJ)	Security-drop rate
LA-CPPO attention	0.9543	52.27	0.1242	0.7372	0.0308
Matched MLP CPPO	0.9228	51.25	0.1554	0.7597	0.0649
Static energy-risk-ETX	0.8807	56.86	0.2392	1.0383	0.1128
Periodic ROA + energy-risk-ETX	0.8814	57.58	0.2332	1.0056	0.1093

Table VII. Exploratory decomposition of the Class-0 total violation difference.

Component	Attention - MLP	Interpretation
Class-0 loss rate	-0.02748	Dominant contributor to lower total violation
Delivered-late Class-0 fraction	-0.00377	Small difference; exploratory CI includes zero
Total Class-0 violation	-0.03125	Confirmatory co-primary effect

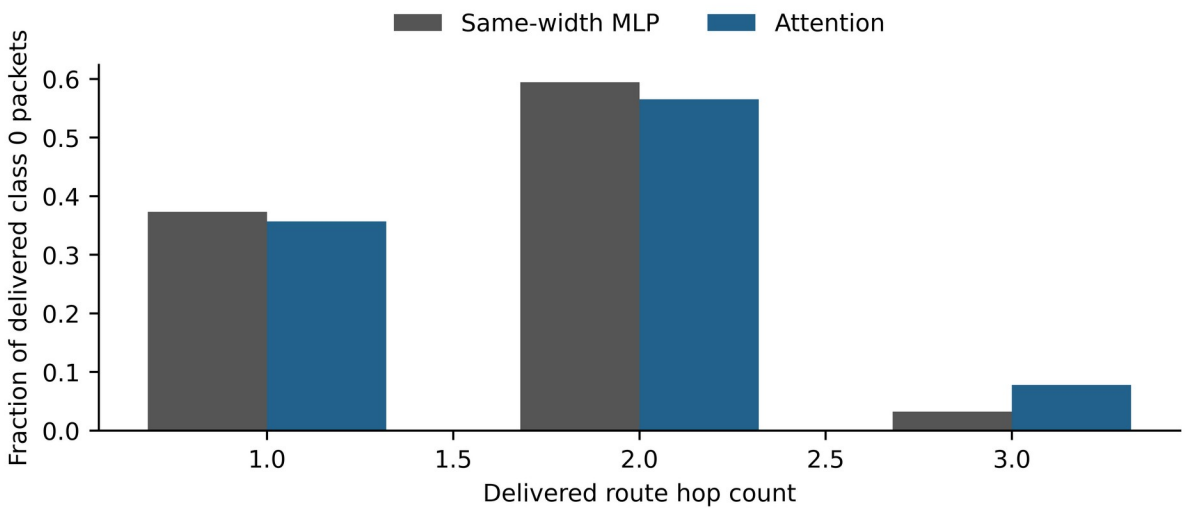


Fig. 5. Pooled hop-count composition of delivered Class 0 packets in the historical experiment.

Table VIII. New nominal validation effects for attention minus each comparator.

Comparator	Endpoint	Difference	95% CI	Holm p
Capacity-matched MLP	Delay ms	+0.821	[-0.121, +1.762]	0.3409
Capacity-matched MLP	Violation pp	-1.522	[-2.754, -0.291]	0.08581
Attention without risk	Delay ms	+0.597	[-0.651, +1.846]	1
Attention without risk	Violation pp	-7.934	[-10.143, -5.725]	2.586e-07
Attention with 80% feedback	Delay ms	-0.258	[-1.068, +0.552]	1
Attention with 80% feedback	Violation pp	-0.518	[-1.761, +0.726]	1

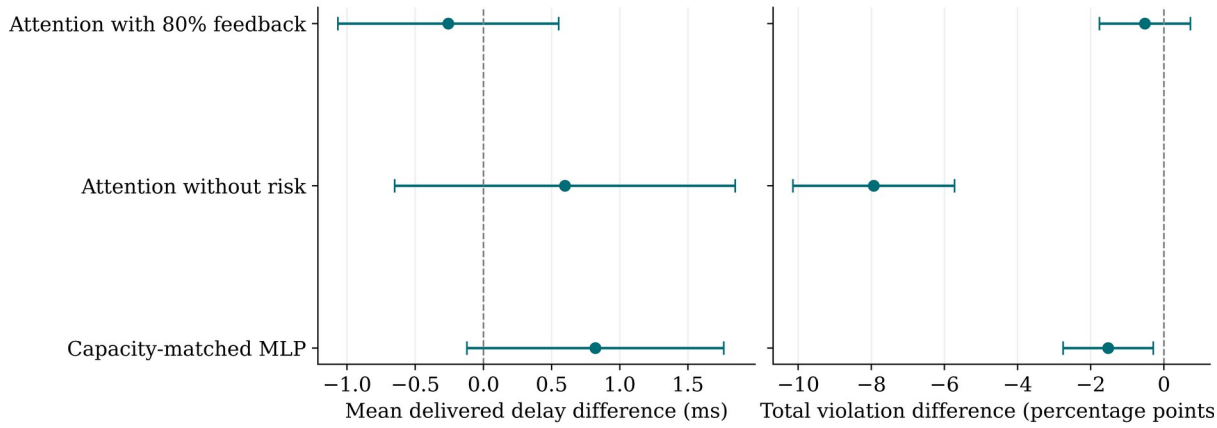


Fig. 6. New nominal paired effects with unadjusted 95% intervals across 30 seeds. Negative differences favor attention.

Table IX. Frozen-policy transfer summaries across 30 seeds.

Setting	Policy	Delivery %	C0 delay ms	C0 violation %
attack10	attention	98.01	49.66	7.36
attack10	capacity_mlp	97.15	48.95	8.13
attack40	attention	93.83	53.46	15.74
attack40	capacity_mlp	89.94	51.67	18.43
nominal	attention	96.92	51.11	9.68
nominal	capacity_mlp	94.96	50.29	11.20
size200	attention	95.92	51.82	11.52
size200	capacity_mlp	94.15	51.26	13.33

Table X. Shared-target UDP diagnostic across 30 paired fixtures.

Policy	**Parameters**	**Executed runs**	**Mean delivery %**
Capacity MLP	63,554	60	32.22
Attention	63,368	60	85.39

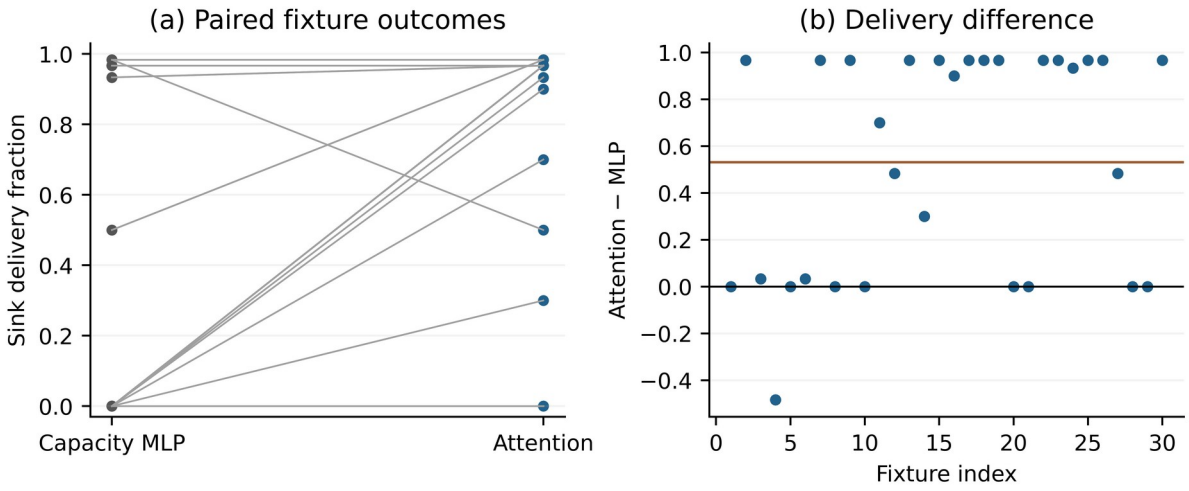


Fig. 7. Shared-target UDP delivery. Each fixture averages two fault-target strata; lines join paired controller outcomes.

Table XI. Historical attention constraint rates across 30 seeds.

Class	Total violation rate	Loss rate	Budget for each channel
0	0.1242	0.0480	0.015
1	0.0444	0.0444	0.04
2	0.0455	0.0455	0.1